SE301 Project 2026-27

Singapore Vertical Farming Research Management System (SVFRMS)
Refactoring & Modernisation: Technical Report

Group: 6

Team members:

Yuen Wai Yee

Jaren Lim Haan

Liew Tian Le

1. Legacy System Analysis

SVFRMS monitors the growing chambers of a vertical-farming facility. Each chamber grows one crop. Sensors report temperature, humidity, soil moisture, light and nutrient levels. A reading outside the crop's range raises an alert, records water or energy use and triggers a corrective device (cooling fan, water pump, humidifier, …). At the end of a run the system archives alerts, prints a daily report and saves an archive report to `archive/daily_report.txt`.

The legacy code is one 393-line class, SVFRMS, whose main components are:

Component	Legacy location
Global mutable state: alert/log/report lists, counters, chamber maps, four locks	`public static` fields, lines 7–31
Chamber state and per-chamber summary	nested classes `ChamberState`, `ChamberSummary`, lines 33–60
Event handling: logging, alert creation, tracking	`handleEvent(String type, …)` string switch, lines 62–88
Device control and its queue (producer–consumer)	`activateControl` if/else chain (90–129); `addControlRequest` + `ControlThread`, `wait/notifyAll` on an `ArrayList` (131–139, 251–273)
Crop threshold rules	`processSensor`: 65 lines of nested `if (crop.equals(...))`, lines 141–205
Reports	`generateDailyReport`, `archiveAlerts`, `saveArchiveReport`, lines 207–241, 309–347
Simulation and start-up	`ChamberThread extends Thread` (275–295), `main` (349–392)

2. Code Analysis and Limitations

2.1 Design issues and code smells

#	Issue (code smell)	Evidence	Principle
D1	**God class** (large class, divergent change): sensing, crop rules, devices, alerts, reports, file I/O, threads and simulation in one class.	whole file	SRP
D2	**Long Method, Duplicated Code and Switch Statements**: adding Kale means another crop branch in up to three sensor sections of `processSensor`; thresholds, costs and devices are **magic numbers** repeated per crop.	lines 145–188	OCP
D3	**Primitive Obsession and Switch Statements**: `handleEvent("TRACK_ALERT", …)`, `activateControl(…, "Cooling Fan")`; queue items are `chamber + "|" + control`, split again later. Typos fail silently; a `|` in a chamber name breaks parsing.	lines 62–129, 136, 269	OCP, type safety
D4	**Refused Bequest** (inheritance used for implementation): `ChamberThread` and `ControlThread` inherit the full `Thread` API although they only need its `run` hook. `main` substitutes them as `Thread`s (`Thread t1 = new ChamberThread(…)`, then `start()/join()`, lines 350–354), and that substitution is LSP-safe: only `run()` is overridden, and `Thread` never promises to stop on `interrupt()`. The cost is design: execution policy is fixed into each task, while `ControlThread` swallowing `InterruptedException` (line 258) is the separate reliability defect D8/C7.	lines 251–275, 350–354	LSP (checked: none), composition over inheritance
D5	**Long Parameter List with a Flag Argument**: `handleEvent(type, chamber, crop, message)` bundles six unrelated operations (log, print, create alert, track alert, track action, …) selected by a string flag; the parameters change meaning per flag and some calls pass `""` for unused ones. It should be split into separate operations, each exposed through a small interface to the client that needs it.	lines 62–88	ISP, SRP
D6	**Dependence on globals and concretions**: every collaborator is a `public static` field; nothing can be replaced or tested alone.	lines 10–31	DIP
D7	**Dead code**: `waterPumpLock`, `nutrientLock`, `irrigationCycle`, `feedingCycle`, `showAlerts`, `showLogs`, the `ALERT_LOG` branch, commented-out `sleep` calls — none is ever called.	lines 28–29, 66–68, 224, 243–249, 297–307, 385	Clean code
D8	**Swallowed exceptions**: `catch(Exception e){}` hides interrupts; `saveArchiveReport` prints an I/O error and the run still "succeeds".	lines 134, 258, 344	Reliability
D9	**Duplicated Code** in reporting: `generateDailyReport` and `saveArchiveReport` build the same header and totals lines separately, so a format change must be made twice.	lines 210–214, 313–318	SRP, DRY
D10	**Hard-coded configuration**: the chamber/crop layout, cycle count (20), queue size (5) and simulated sensor ranges are literals in `main` and the threads; any change means editing and recompiling.	lines 15, 285–290, 351–354	OCP

2.2 Behavioural quirks (kept, see §3.3)

- Soil checks exist only for Tomato (and only its minimum); humidity checks only for Tomato and Strawberry. The spec's soil range for Strawberry and humidity/soil ranges for Lettuce were never coded.
- Light (< 500 lux) and nutrient (< 40 %) limits are facility-wide, and their alert text omits the crop. Low temperature alerts but starts no device (no Heating Unit).
- A queued device request can run after a newer reading of the same sensor (a fan queued at 35 °C runs after a 20 °C reading). This is a logical ordering issue, not a data race.

- Never implemented: severity levels, Sensor Failure alerts, Heating Unit, Ventilation System, Shade Controller and the weekly, water, energy and nutrient reports. §5 shows where each plugs in; severity and Sensor Failure were added (§3.3).

2.3 Concurrency issues

Four `ChamberThreads` and one `ControlThread` share all static state with no consistent locking.

#	Type	Where	Failure
C1	Race condition	`alerts` and `sensorLogs` are plain `ArrayLists` written by the 4 chamber threads (lines 64, 73)	Lost alerts, corrupted array, `ArrayIndexOutOfBoundsException`
C2	Check-then-act race	`getChamberState`/`getChamberSummary`: `containsKey` → `put` on a `HashMap`	Two states for one chamber; `HashMap` corruption → `NullPointerException`
C3	Lost update	`totalWaterUsage += …`, `totalEnergyUsage`, `totalAlerts++`, `totalAlertsGenerated++`	Non-atomic read-modify-write → wrong totals
C4	Data race / visibility	`ChamberState` fields (lines 33–38) are written by chamber threads (lines 141–205) and read-modify-written by the control thread (lines 91–128) with no lock	Device effect lost or computed from a stale value
C5	Lost summary (via C2)	`getChamberSummary` is called by a chamber thread (alerts) and the control thread (actions) for the same chamber; its check-then-act can create two `ChamberSummary` objects, and one is lost	Summary alert/action counts don't add up
C6	Order race	`main` starts `archiveAlerts` and `generateDailyReport` together (lines 373–389); both take the alert and report locks (lines 207–240)	The two console report blocks swap order, "Active Alerts" is 0 or N at random, and the archive's activity lines ("Daily Report", "Archived N alerts") also swap (the commented `sleep(20)` shows it was noticed)
C7	Liveness (deadlock-like hang)	Producers `wait()` on a full queue with no timeout; if `ControlThread` dies (any `RuntimeException`, e.g. the C2 NPE) nobody consumes	All chamber threads block forever; `main` hangs in `join()`
C8	Latent deadlock	Nested `synchronized(alertLock){ synchronized(reportLock){…} }` in two methods (lines 208–209, 223–230); the locks are `public static`, **non-final**	No deadlock today, because both methods take the locks in the same order (alert, then report), and `wait()` releases the queue monitor. But any new code that takes them in the other order, or reassigns a lock, deadlocks or breaks mutual exclusion
C9	Unsafe iteration (latent)	`showAlerts`/`showLogs` iterate lists other threads modify; both are dead code today (D7)	`ConcurrentModificationException` as soon as either is called during a run

Measured (150 runs per version, Linux, JDK 21; each run checked against its own synchronized console output). The rates depend on the platform: on Windows 11 with JDK 25 the legacy version crashed with a `NullPointerException` in 42 of 60 runs and the refactored version in none; `tools/compare_legacy.sh` reproduces the count and the output comparison of §3.3.

Symptom	Legacy	Refactored
Crash: `NullPointerException` in a chamber thread (C2)	3	0
Archive lists fewer alerts than were raised (C1)	5	0
Wrong water/energy total (C3) / wrong "Generated Alerts" (C3)	5 / 2	0 / 0
Chamber summary counts wrong (C5)	1	0
"Active Alerts" in the daily report (C6)	0 in 130 runs, N in 20	0 in all 150

No hang (C7) occurred in these runs because the NPEs hit chamber threads. `legacy/LegacyHangDemo.java` forces the same NPE in the control thread, by swapping the public, non-final `chamberStates` field (D6) without editing the legacy code. It hangs in 5 of 5 runs: all four chamber threads wait forever in `addControlRequest` (line 134) and `main` never returns. Refactored, a dead worker wakes and rejects blocked producers (`ControlDispatcherLifecycleTest`), and `SVFRMS.run` turns the failed run into exit status 1 with no report (`SimulationRunnerTest`).

3. Proposed Design

3.1 Key decisions, by SOLID principle

Principle	Change (fixes)
SRP (D1)	Nine packages plus the composition root, one reason to change each (§4). `ControlDispatcher` owns worker lifecycle, `ClosableBoundedQueue` blocking, and injected `RequestExecutor` command execution. Reporting separates orchestration (`ReportService`), activity/snapshot state (`FacilitySnapshotFactory`) and persistence (`ReportWriter`).
OCP (D2, D3)	Crops are data: `SensorProcessor` has one generic *breach → alert → cost → device* rule, so crops are added or removed without editing it. `ReportService.print/save` accepts any `Report`, and `ResourceMeter` initializes an `EnumMap` from every `Resource`, so new reports and resource kinds do not add type branches. Every "for each sensor" loop iterates `SensorType`.

Principle	Change (fixes)
LSP (D4)	Composition replaces `Thread` inheritance: chamber work is a `Callable` run by an `ExecutorService`, and the dispatcher owns its worker. The new design has real substitution points, and the tests use them: scripted `ReadingSources` (`SensorExtensibilityTest`), throwing and blocking `Devices` (`ControlDispatcherLifecycleTest`), lambda `DeviceControllers` and failing listeners (`ListenerIsolationTest`). The code under test cannot tell a stand-in from the production class.
ISP	Small, client-specific roles (`AlertListener`, `DeviceListener`, `DeviceController`, `Device`, `ChamberControl`, `Report`, `ReadingSource`): each client depends only on what it uses. Services expose a separate interface per client: `AlertService` is an `AlertSink` to monitoring, an `AlertArchive` to the archiving step and an `AlertHistory` to report snapshots; `ResourceMeter` is a `UsageRecorder` to monitoring and a one-method `UsageTotals` to snapshots; `ChamberRegistry` is only a `ReadingRecorder` to monitoring and only `ChamberControls` to devices, so neither can do the other's job.
DIP (D6)	Constructor injection: the composition root chooses the reports, writer, snapshot roles, request executor and error sinks. `ControlDispatcher` receives a `RequestHandler`; publishers never write to global stderr directly. Services build only private data structures such as the queue and listener list. `ArchitectureTest` checks that service constructors take interfaces or immutable values.

Dead code was removed (D7), including the sensor log that legacy wrote but never read; no production method exists only for tests. I/O errors now set the exit status, and interrupts are restored (D8).

3.2 Concurrency fixes

Issue	Fix
C1, C9	Each shared list has one owner that guards it (`AlertService` synchronized). Readers get immutable copies, so iteration never races. The report activity log is written only by the main thread; its `synchronizedList` is defensive.
C2	`ConcurrentHashMap.computeIfAbsent` (atomic get-or-create) in `ChamberRegistry` and `ChamberSummaryTracker`.
C3, C5	`AtomicInteger` counters in `ResourceMeter`; alert counters change under the same lock as the alert list; one summary per chamber (`computeIfAbsent`, C2), updated under its own lock.
C4	`ChamberState` is an `EnumMap` behind `synchronized` methods; every device change is one atomic `adjust` (read, add, clamp, write), so no effect is lost. Device code runs outside the lock (`deviceCodeRunsWithNoChamberLockHeld`). The ordering quirk of §2.2 is functionality, not a race, so it is kept.
C6	Archive, then daily report, sequentially, after all producers and the consumer have finished.
C7	`ClosableBoundedQueue` (one `ReentrantLock`, two `Conditions`) replaces `wait/notify`. Worker death calls `abort`, waking every blocked producer. A full queue is also bounded by the dispatcher's timeout: expiry aborts the queue, interrupts the worker and fails every chamber, so `SimulationRunner` can reach shutdown even when a device never returns. Requests are accepted only in `RUNNING`; close races and close-before-start reject safely. `close()` drains normally, then interrupts after 30 s and allows 500 ms grace; all closers report the same outcome. Queued requests are counted and dropped on timeout; an already-running device may ignore interruption, but its daemon worker cannot hold the application alive.
C8	**No nested locks anywhere**, by construction: each component guards its own state with one lock, and listeners and device code run outside every lock. Reports read an immutable `FacilitySnapshot`, built after every thread has finished.

Tests include exact counters; close racing eight producers; worker death with a full queue; a stuck device with blocked producers; close before start; concurrent closers; and interruption. Mutation-targeted stress tests prove concurrent first access publishes one chamber state, atomic device adjustments lose no updates, and alert/action summary counts remain exact.

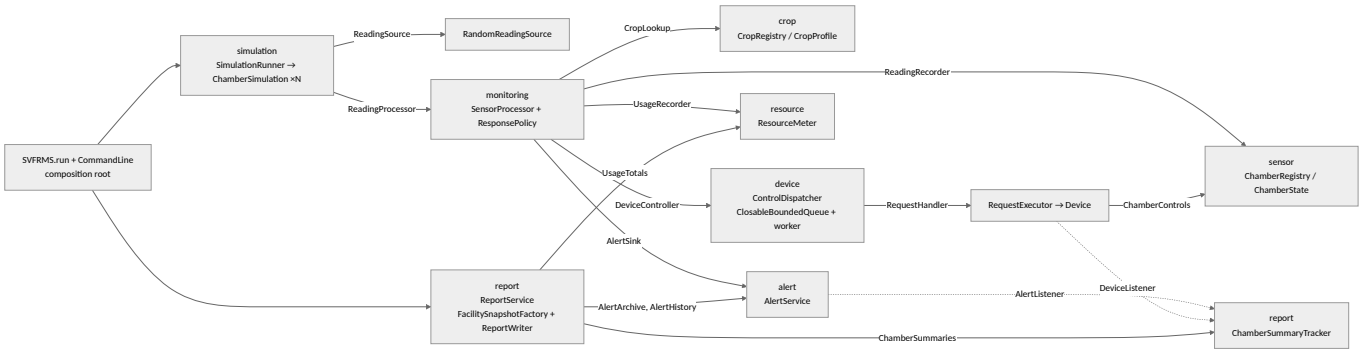
3.3 Behaviour preservation

Thresholds, alert texts, resource costs, device effects, console formats and the archive layout are identical: the sets of distinct output lines of 25 legacy and 25 refactored runs (numbers masked) were diffed and match (`tools/compare_legacy.sh`); `GoldenOutputTest` pins the scripted Tomato run's distinct console lines, exact totals and archive layout in the build. The quirks in §2.2 are kept because the brief forbids changing functionality; each is now a one-line data change (e.g. `.soilMoisture(50, 65)` on Strawberry). The one bound legacy ignores, Tomato soil above 60 %, is declared on the crop (`.ignoreBreach(SOIL_MOISTURE, HIGH)`) rather than hidden in the policy, which must cover every sensor and direction. Deliberate changes appear only when triggered:

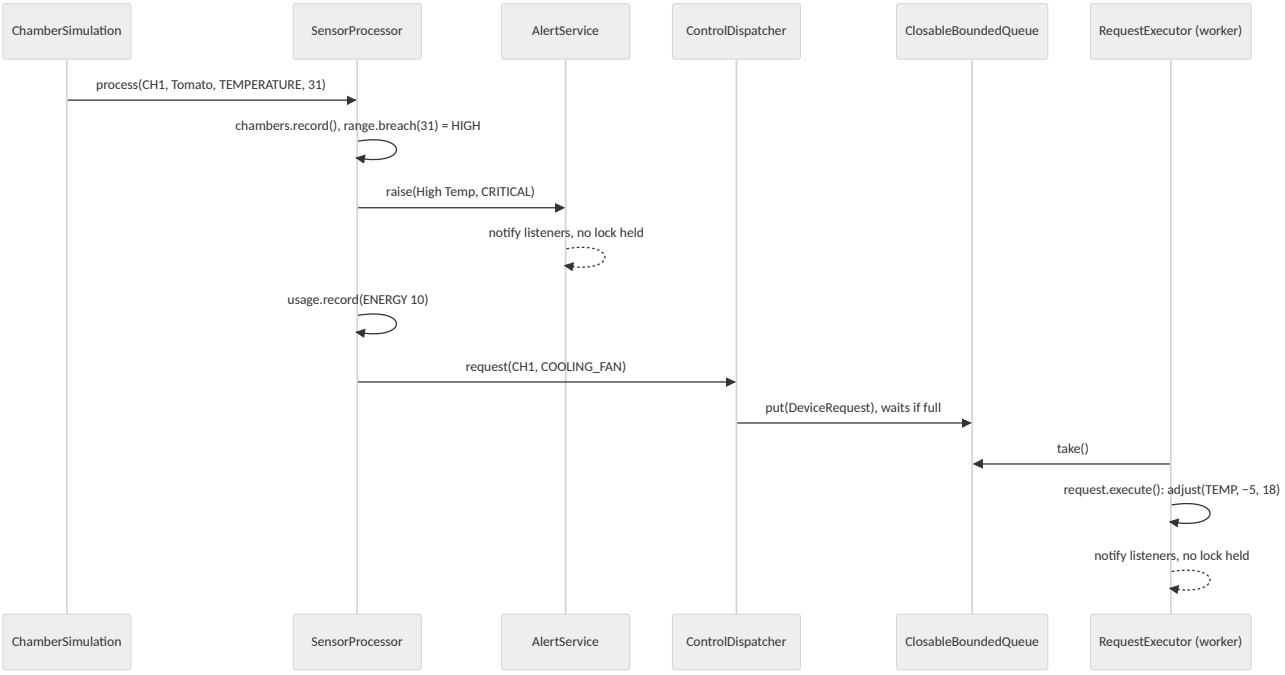
- C6 is deterministic: "Active Alerts" is always 0, the legacy majority outcome.
- **Sensor Failure** (spec §3.4): a reading a sensor cannot produce (NaN, infinite, outside its physical range, e.g. 150 % humidity) or no reading at all raises a CRITICAL `CHn Sensor Failure: …` alert and is never stored or acted on. Ranges, costs and resource amounts are validated. The random feed never fails, so this path is reached through the `ReadingSource` seam (tests, real hardware).
- **Severity** (spec §3.4): every alert carries INFO, WARNING or CRITICAL (temperature breaches are CRITICAL, as in the spec's example). `--alert-format=severity` prints `CRITICAL: CH3 Lettuce High Temp`; the default console format is unchanged.
- `CHn=Crop` arguments replace the default four chambers (`java -jar run.jar CH1=Tomato CH5=Kale` runs two). New crops get alerts on both bounds of every range they define (e.g. `CH5 Kale High Soil Moisture`).
- If the archive cannot be saved, the legacy message is printed and the cause goes to stderr.
- Exit status: 0 success; 1 run failed (no report written); 2 archive not saved; 3 invalid arguments.

4. Final Architecture

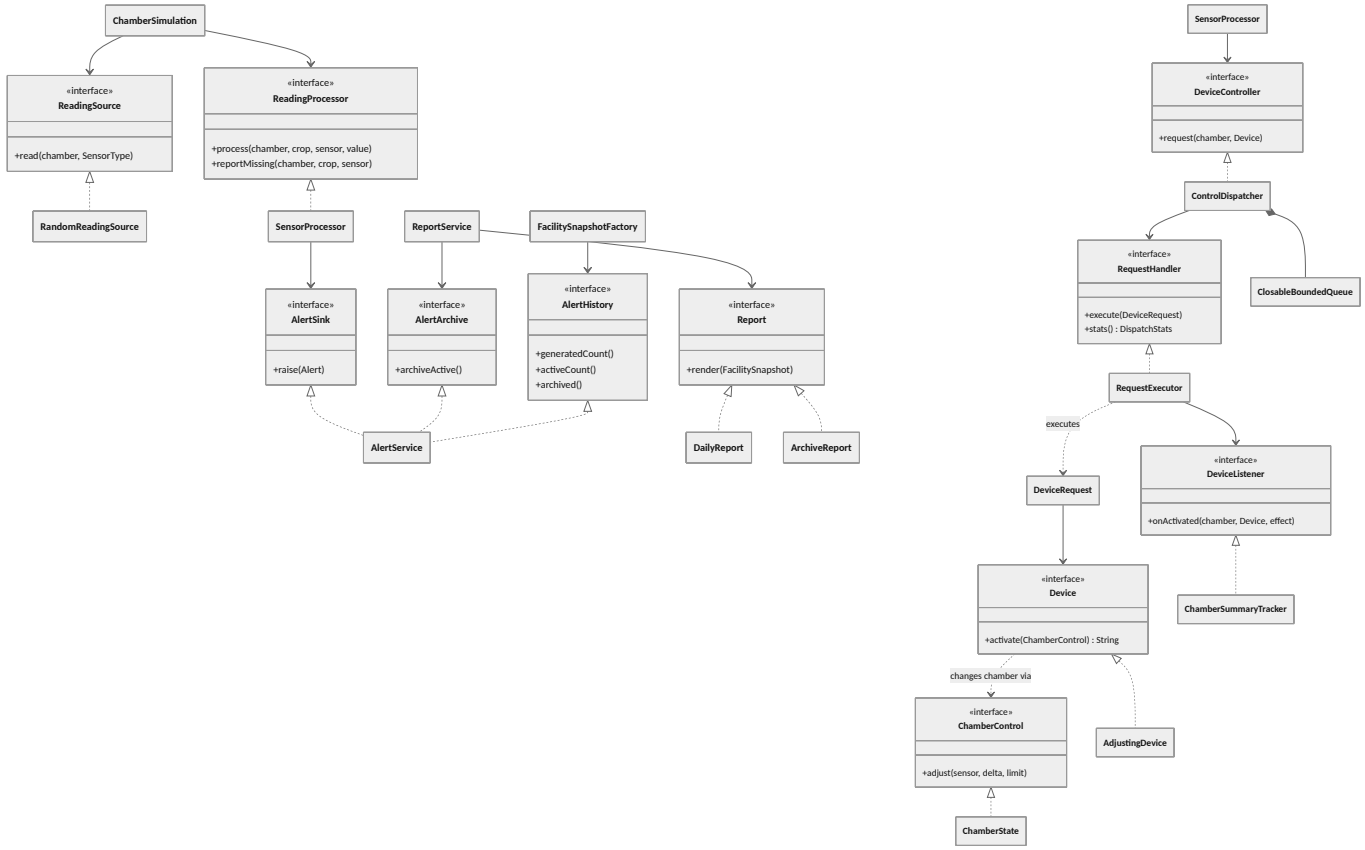
Component diagram



Sequence: one out-of-range reading



Class diagrams: monitoring, alerts and reporting (left); device control (right)



Package	Main classes	Responsibility
`crop`	`CropProfile` (+Builder), `Range`, `CropRegistry`, `CropCatalog`, `CropSpecParser`	What each crop needs
`sensor`	`SensorType`, `Breach`, `ChamberState`, `ChamberRegistry` (`ReadingRecorder`, `ChamberControls`)	Readings and chamber conditions
`monitoring`	`SensorProcessor`, `ResponsePolicy`, `Response`	Reading → alert / cost / device rule
`device`	`ControlDispatcher`, `ClosableBoundedQueue`, `RequestExecutor`, `DeviceRequest`, `Devices`	Executing corrective actions
`alert`, `resource`	`AlertService`, `Severity`, printers; `ResourceMeter`, `Usage`	Alert lifecycle; water and energy
`report`	`ReportService`, `FacilitySnapshotFactory`, `ReportWriter`, reports, `ChamberSummaryTracker`	Report orchestration, snapshots and persistence
`simulation`, `support`	`ChamberSimulation`, `SimulationRunner`, `RandomReadingSource`; `Listeners`	Sensor feed; isolated notification

5. Extensibility and Maintainability

Change	What to touch
Add / remove a crop	One `CropCatalog` entry, or no code at all: `--add-crop="Okra:temperature=24..30,soil=50..70"`, `--remove-crop=Tomato`.
Add a device (e.g. Heating Unit)	One `Devices` constant, referenced from the `Low Temp` entry of `ResponsePolicy.standard()`; no logic changes.
New sensor type	One `SensorType` constant, its two `ResponsePolicy` entries and its simulated range; start-up fails fast if either is missing. Parsing, simulation and state follow automatically.
New report (weekly, water, energy, nutrient)	A new `Report` rendering `FacilitySnapshot`, then one composition-root call to generic `ReportService.print` or `save`; `ReportService` is unchanged. New `Resource` enum values are metered automatically and available through `UsageTotals.totals()` and `FacilitySnapshot.resourceUsage()`.
New alert channel or severity routing	A new `AlertListener` registered in `SVFRMS.run`; a runtime failure is isolated and cannot disturb the others. A new console format is one entry in the `AlertFormats` table; validation, the usage text and wiring all read that table.
Real sensors	A new `ReadingSource`. A source that returns no reading raises a Sensor Failure alert; one that blocks is stopped by the run timeout in `SimulationRunner`.

Every change is additive and local. Constructor injection lets each component be tested alone: 76 tests in 15 classes cover behaviour parity, every crop bound, validation, concurrency, shutdown, run-time crops and exit codes.

6. Design Patterns

Pattern	Where	Problem solved	Why this rather than the alternative
Builder	`CropProfile.Builder`	Replaces legacy thresholds, costs and exceptions scattered as magic numbers across crop branches with one validated immutable profile.	A telescoping constructor would mix many optional ranges/costs; mutable setters would expose partially built profiles to chamber threads.
Strategy	`Report` (`DailyReport`, `ArchiveReport`) injected into `ReportService`; `ReadingSource` (`RandomReadingSource`, test sources) injected into `ChamberSimulation`	Replaces legacy hard-wired report routines and the hard-wired `Random` feed with interchangeable algorithms chosen by the composition root.	Type flags would recreate `handleEvent`-style switches; subclassing the simulation or report service would couple algorithm choice to lifecycle code.
Observer	`AlertService` → `AlertListeners`; `RequestExecutor` → `DeviceListeners`	Producers must not know who consumes alerts (console, summary, future email).	Direct calls (`handleEvent`) couple alerting to every consumer; fault isolation in `Listeners` keeps one bad listener from stopping the rest.
Producer–Consumer with guarded suspension	`ControlDispatcher` + `ClosableBoundedQueue`	Back-pressure, one writer for device effects, clean and abnormal shutdown.	`ArrayBlockingQueue` + poison pill (our first version) cannot wake producers when the consumer dies. A bounded `ThreadPoolExecutor` rejects new work when full instead of blocking the producer, so it loses legacy back-pressure.
Special Case	`SensorProcessor`: an unknown crop gets the default `CropProfile` produced by `CropProfile.builder("Unknown").build()`, built once and shared (facility light and nutrient limits only)	Unknown crops still get the facility-wide checks, as in legacy.	A missing profile would require `null` checks around every range lookup.

Supporting idioms: **Request object** (`DeviceRequest`: a typed chamber + `Device` pair, replacing the legacy `"chamber|control"` strings (D3), executed later by the control worker); **Static factory methods** (`CropCatalog.defaultRegistry()`, `ResponsePolicy.standard()`, `RandomReadingSource.legacy()`) name the legacy configurations; **Registry** (`CropRegistry`, `ChamberRegistry`, thread-safe lookup); **Immutable Value Objects** (`FacilitySnapshot`, `CropProfile`, records), so reports never hold a lock; **Dependency Injection** with a single composition root (`SVFRMS.run`).

7. Build and Run

`mvn clean package` builds `target/run.jar` and runs the 76 tests. `java -jar run.jar` runs the legacy four chambers; `java -jar run.jar --help` lists the options.